

OSSP (Operating Systems And Systems Programming)

Course Code: 25CS2104E

Section: 3

Project: Database Client-Server Management System

Project Output Screenshots

Terminal 1(Server): -

Open Terminal 1

```
aswitha_patel@Aswitha:~$ cd database_client_server
aswitha_patel@Aswitha:~/database_client_server$ tree
```

```
├── Makefile
├── README.md
├── client
├── config.txt
├── data
│   └── students.dat
├── include
│   ├── common.h
│   ├── config.h
│   ├── database.h
│   ├── logger.h
│   ├── monitor.h
│   ├── queue.h
│   └── server.c
├── logs
│   ├── server.log
│   └── transaction.log
├── server
├── src
│   ├── client.c
│   ├── config.c
│   ├── config.h
│   ├── database.c
│   ├── logger.c
│   ├── monitor.c
│   ├── queue.c
│   ├── server.c
│   ├── test_database.c
│   ├── test_logger.c
│   └── test_queue.c
├── test_database
├── test_logger
└── test_queue
```

5 directories, 29 files

```
aswitha_patel@Aswitha:~/database_client_server$ |
```

Compile the project

```
aswitha_patel@Aswitha:~/database_client_server$ make clean
rm -f server client
aswitha_patel@Aswitha:~/database_client_server$ |
```

```
aswitha_patel@Aswitha:~/database_client_server$ make
gcc -Wall -Wextra -Iinclude -o server src/server.c src/database.c src/queue.c src/logger.c src/config.c src/monitor.c -pthread -lrt
gcc -Wall -Wextra -Iinclude -o client src/client.c -pthread -lrt
aswitha_patel@Aswitha:~/database_client_server$ |
```

```
aswitha_patel@Aswitha:~/database_client_server$ ls -lh server client
-rwxr-xr-x 1 aswitha_patel aswitha_patel 17K Sep 23 16:25 client
-rwxr-xr-x 1 aswitha_patel aswitha_patel 40K Sep 23 16:25 server
aswitha_patel@Aswitha:~/database_client_server$ |
```

Start the server

```
aswitha_patel@Aswitha:~/database_client_server$ ./server
```

```
===== SERVER CONFIGURATION =====
Port           : 8080
Workers        : 4
Max Clients    : 20
Log Level      : 1
=====

=====
Database Client-Server Management System
=====
Server running on port 8080
Maximum clients: 20
Worker threads: 4
Waiting for client connections...
|
```

Terminal 2 (Client): -

```
aswitha_patel@Aswitha:~$ cd database_client_server
aswitha_patel@Aswitha:~/database_client_server$ |
```

```
aswitha_patel@Aswitha:~/database_client_server$ ./client
Connected to Database Server!
Server IP: 127.0.0.1
Server Port: 8080

Available commands:
INSERT id name course marks attendance
SEARCH id
UPDATE id marks attendance
DELETE id
DISPLAY
MONITOR
EXIT

Enter request: |
```

Demonstrate CRUD

Insert Operation

```
aswitha_patel@Aswitha:~/database_client_server$ ./client
Connected to Database Server!
Server IP: 127.0.0.1
Server Port: 8080

Available commands:
INSERT id name course marks attendance
SEARCH id
UPDATE id marks attendance
DELETE id
DISPLAY
MONITOR
EXIT

Enter request: INSERT 30 Ravi CSE 85 90
Server response: SUCCESS: Student inserted

Enter request: |
```

Search Operation

```
aswitha_patel@Aswitha:~/database_client_server$ ./client
Connected to Database Server!
Server IP: 127.0.0.1
Server Port: 8080

Available commands:
INSERT id name course marks attendance
SEARCH id
UPDATE id marks attendance
DELETE id
DISPLAY
MONITOR
EXIT

Enter request: INSERT 30 Ravi CSE 85 90
Server response: SUCCESS: Student inserted

Enter request: SEARCH 30
Server response: SUCCESS: ID=30 Name=Ravi Course=CSE Marks=85.00 Attendance=90.00

Enter request: |
```

Update operation

```
aswitha_patel@Aswitha:~/database_client_server$ ./client
Connected to Database Server!
Server IP: 127.0.0.1
Server Port: 8080

Available commands:
INSERT id name course marks attendance
SEARCH id
UPDATE id marks attendance
DELETE id
DISPLAY
MONITOR
EXIT

Enter request: INSERT 30 Ravi CSE 85 90
Server response: SUCCESS: Student inserted

Enter request: SEARCH 30
Server response: SUCCESS: ID=30 Name=Ravi Course=CSE Marks=85.00 Attendance=90.00

Enter request: UPDATE 30 95 95
Server response: SUCCESS: Student updated

Enter request: |
```

Searching after updating

```

aswitha_patel@Aswitha:~$ cd database_client_server
aswitha_patel@Aswitha:~/database_client_server$ ./client
Connected to Database Server!
Server IP: 127.0.0.1
Server Port: 8080

Available commands:
INSERT id name course marks attendance
SEARCH id
UPDATE id marks attendance
DELETE id
DISPLAY
MONITOR
EXIT

Enter request: INSERT 30 Ravi CSE 85 90
Server response: SUCCESS: Student inserted

Enter request: SEARCH 30
Server response: SUCCESS: ID=30 Name=Ravi Course=CSE Marks=85.00 Attendance=90.00

Enter request: UPDATE 30 95 95
Server response: SUCCESS: Student updated

Enter request: SEARCH 30
Server response: SUCCESS: ID=30 Name=Ravi Course=CSE Marks=95.00 Attendance=95.00

Enter request: |

```

Display operation

```

aswitha_patel@Aswitha:~/database_client_server$ ./client
Connected to Database Server!
Server IP: 127.0.0.1
Server Port: 8080

Available commands:
INSERT id name course marks attendance
SEARCH id
UPDATE id marks attendance
DELETE id
DISPLAY
MONITOR
EXIT

Enter request: INSERT 30 Ravi CSE 85 90
Server response: SUCCESS: Student inserted

Enter request: SEARCH 30
Server response: SUCCESS: ID=30 Name=Ravi Course=CSE Marks=85.00 Attendance=90.00

Enter request: UPDATE 30 95 95
Server response: SUCCESS: Student updated

Enter request: SEARCH 30
Server response: SUCCESS: ID=30 Name=Ravi Course=CSE Marks=95.00 Attendance=95.00

Enter request: DISPLAY
Server response: SUCCESS: Student records displayed on server terminal

Enter request: |

```

```

aswitha_patel@Aswitha:~/database_client_server$ ./server

===== SERVER CONFIGURATION =====
Port      : 8080
Workers   : 4
Max Clients : 20
Log Level  : 1
=====

=====
Database Client-Server Management System
=====
Server running on port 8080
Maximum clients: 20
Worker threads: 4
Waiting for client connections...

===== STUDENT DATABASE =====
ID: 1 | Name: Aswitha | Course: CSE | Marks: 92.00 | Attendance: 95.00%
ID: 20 | Name: Alice | Course: CSE | Marks: 88.00 | Attendance: 90.00%
ID: 11 | Name: StudentB | Course: ECE | Marks: 75.00 | Attendance: 90.00%
ID: 21 | Name: Bob | Course: ECE | Marks: 92.00 | Attendance: 95.00%
ID: -1774936961 | Name: Manasvi | Course: CSE | Marks: 99.00 | Attendance: 89.00%
ID: 6 | Name: Amulya | Course: CSE | Marks: 99.00 | Attendance: 89.00%
ID: 30 | Name: Ravi | Course: CSE | Marks: 95.00 | Attendance: 95.00%
=====

```

Delete operation

```

Enter request: DELETE 30
Server response: SUCCESS: Student deleted

Enter request: |

```

Searching after deleting

```

Enter request: DELETE 30
Server response: SUCCESS: Student deleted

Enter request: SEARCH 30
Server response: ERROR: Student not found

Enter request: |

```

Terminal 3 (Client 2): -

Demonstrate multiple clients

```
aswitha_patel@Aswitha:~$ cd database_client_server
aswitha_patel@Aswitha:~/database_client_server$ ./client
Connected to Database Server!
Server IP: 127.0.0.1
Server Port: 8080

Available commands:
INSERT id name course marks attendance
SEARCH id
UPDATE id marks attendance
DELETE id
DISPLAY
MONITOR
EXIT

Enter request: |
```

```
Connected to Database Server!
Server IP: 127.0.0.1
Server Port: 8080

Available commands:
INSERT id name course marks attendance
SEARCH id
UPDATE id marks attendance
DELETE id
DISPLAY
MONITOR
EXIT

Enter request: INSERT 19 Akshaya CSE 88 90
Server response: SUCCESS: Student inserted

Enter request: INSERT 22 Srujana ECE 92 95
Server response: SUCCESS: Student inserted

Enter request: |
```

```

aswitha_patel@Aswitha:~$ cd database_client_server
aswitha_patel@Aswitha:~/database_client_server$ ./client
Connected to Database Server!
Server IP: 127.0.0.1
Server Port: 8080

Available commands:
INSERT id name course marks attendance
SEARCH id
UPDATE id marks attendance
DELETE id
DISPLAY
MONITOR
EXIT

Enter request: INSERT 19 Akshaya CSE 88 90
Server response: SUCCESS: Student inserted

Enter request: INSERT 22 Srujana ECE 92 95
Server response: SUCCESS: Student inserted

Enter request: SEARCH 20
Server response: SUCCESS: ID=20 Name=Alice Course=CSE Marks=88.00 Attendance=90.00

Enter request: SEARCH 22
Server response: SUCCESS: ID=22 Name=Srujana Course=ECE Marks=92.00 Attendance=95.00

Enter request: SEARCH 19
Server response: SUCCESS: ID=19 Name=Akshaya Course=CSE Marks=88.00 Attendance=90.00

Enter request: |

```

Demonstrate Monitoring: -

In any one of the client terminals

```

Enter request: MONITOR
Server response: SUCCESS: Monitoring information displayed on server terminal

Enter request: |

```

In Server Terminal: -

```

===== SERVER MONITOR =====
Server PID: 1369
VmRSS:      2324 kB
Threads:      8
=====
|

```

Memory mapping

Terminal 4: -

```

aswitha_patel@Aswitha:~$ cd ~/database_client_server
aswitha_patel@Aswitha:~/database_client_server$ grep -n "mmap\|munmap" src/database.c
39:     db = mmap(
49:         perror("mmap failed");
72:         munmap(
aswitha_patel@Aswitha:~/database_client_server$ |

```

Mutexes and Semaphores

```

aswitha_patel@Aswitha:~/database_client_server$ grep -n "pthread_mutex\|sem_" src/database.c src/queue.c
src/database.c:17:pthread_mutex_t db_mutex = PTHREAD_MUTEX_INITIALIZER;
src/database.c:92:     pthread_mutex_lock(&db_mutex);
src/database.c:97:         pthread_mutex_unlock(&db_mutex);
src/database.c:121:         pthread_mutex_unlock(&db_mutex);
src/database.c:133:     pthread_mutex_unlock(&db_mutex);
src/database.c:152:     pthread_mutex_lock(&db_mutex);
src/database.c:159:         pthread_mutex_unlock(&db_mutex);
src/database.c:171:     pthread_mutex_unlock(&db_mutex);
src/database.c:187:     pthread_mutex_lock(&db_mutex);
src/database.c:202:         pthread_mutex_unlock(&db_mutex);
src/database.c:214:     pthread_mutex_unlock(&db_mutex);
src/database.c:229:     pthread_mutex_lock(&db_mutex);
src/database.c:242:         pthread_mutex_unlock(&db_mutex);
src/database.c:254:     pthread_mutex_unlock(&db_mutex);
src/database.c:269:     pthread_mutex_lock(&db_mutex);
src/database.c:297:     pthread_mutex_unlock(&db_mutex);
src/queue.c:16:static pthread_mutex_t request_mutex;
src/queue.c:17:static sem_t request_empty;
src/queue.c:18:static sem_t request_full;
src/queue.c:27:static pthread_mutex_t response_mutex;
src/queue.c:28:static sem_t response_empty;
src/queue.c:29:static sem_t response_full;
src/queue.c:38:     if (pthread_mutex_init(&request_mutex, NULL) != 0) {
src/queue.c:43:         if (sem_init(&request_empty, 0, QUEUE_SIZE) != 0) {

```

```

src/queue.c:45:         pthread_mutex_destroy(&request_mutex);
src/queue.c:49:         if (sem_init(&request_full, 0, 0) != 0) {
src/queue.c:51:             sem_destroy(&request_empty);
src/queue.c:52:             pthread_mutex_destroy(&request_mutex);
src/queue.c:63:             sem_destroy(&request_empty);
src/queue.c:64:             sem_destroy(&request_full);
src/queue.c:65:             pthread_mutex_destroy(&request_mutex);
src/queue.c:72:             if (sem_wait(&request_empty) != 0) {
src/queue.c:77:                 pthread_mutex_lock(&request_mutex);
src/queue.c:82:                 pthread_mutex_unlock(&request_mutex);
src/queue.c:84:                 if (sem_post(&request_full) != 0) {
src/queue.c:100:                     if (sem_wait(&request_full) != 0) {
src/queue.c:105:                         pthread_mutex_lock(&request_mutex);
src/queue.c:110:                         pthread_mutex_unlock(&request_mutex);
src/queue.c:112:                         if (sem_post(&request_empty) != 0) {
src/queue.c:127:                             if (pthread_mutex_init(&response_mutex, NULL) != 0) {
src/queue.c:132:                                 if (sem_init(&response_empty, 0, QUEUE_SIZE) != 0) {
src/queue.c:134:                                     pthread_mutex_destroy(&response_mutex);
src/queue.c:138:                                     if (sem_init(&response_full, 0, 0) != 0) {
src/queue.c:140:                                         sem_destroy(&response_empty);
src/queue.c:141:                                         pthread_mutex_destroy(&response_mutex);
src/queue.c:152:                                         sem_destroy(&response_empty);
src/queue.c:153:                                         sem_destroy(&response_full);
src/queue.c:154:                                         pthread_mutex_destroy(&response_mutex);
src/queue.c:161:                                         if (sem_wait(&response_empty) != 0) {
src/queue.c:166:                                             pthread_mutex_lock(&response_mutex);
src/queue.c:171:                                             pthread_mutex_unlock(&response_mutex);
src/queue.c:173:                                             if (sem_post(&response_full) != 0) {
src/queue.c:189:                                                 if (sem_wait(&response_full) != 0) {

```

```

src/queue.c:52:         pthread_mutex_destroy(&request_mutex);
src/queue.c:63:     sem_destroy(&request_empty);
src/queue.c:64:     sem_destroy(&request_full);
src/queue.c:65:     pthread_mutex_destroy(&request_mutex);
src/queue.c:72:     if (sem_wait(&request_empty) != 0) {
src/queue.c:77:     pthread_mutex_lock(&request_mutex);
src/queue.c:82:     pthread_mutex_unlock(&request_mutex);
src/queue.c:84:     if (sem_post(&request_full) != 0) {
src/queue.c:100:     if (sem_wait(&request_full) != 0) {
src/queue.c:105:     pthread_mutex_lock(&request_mutex);
src/queue.c:110:     pthread_mutex_unlock(&request_mutex);
src/queue.c:112:     if (sem_post(&request_empty) != 0) {
src/queue.c:127:     if (pthread_mutex_init(&response_mutex, NULL) != 0) {
src/queue.c:132:     if (sem_init(&response_empty, 0, QUEUE_SIZE) != 0) {
src/queue.c:134:         pthread_mutex_destroy(&response_mutex);
src/queue.c:138:     if (sem_init(&response_full, 0, 0) != 0) {
src/queue.c:140:         sem_destroy(&response_empty);
src/queue.c:141:         pthread_mutex_destroy(&response_mutex);
src/queue.c:152:     sem_destroy(&response_empty);
src/queue.c:153:     sem_destroy(&response_full);
src/queue.c:154:     pthread_mutex_destroy(&response_mutex);
src/queue.c:161:     if (sem_wait(&response_empty) != 0) {
src/queue.c:166:     pthread_mutex_lock(&response_mutex);
src/queue.c:171:     pthread_mutex_unlock(&response_mutex);
src/queue.c:173:     if (sem_post(&response_full) != 0) {
src/queue.c:189:     if (sem_wait(&response_full) != 0) {
src/queue.c:194:     pthread_mutex_lock(&response_mutex);
src/queue.c:199:     pthread_mutex_unlock(&response_mutex);
src/queue.c:201:     if (sem_post(&response_empty) != 0) {
aswitha_patel@Aswitha:~/database_client_server$ |

```

Logging

```

aswitha_patel@Aswitha:~/database_client_server$ ls logs
server.log  transaction.log
aswitha_patel@Aswitha:~/database_client_server$ |

```

```

aswitha_patel@Aswitha:~/database_client_server$ cat logs/server.log
[2026-09-17 09:28:14] SERVER: Logger initialized successfully
[2026-09-17 09:28:14] SERVER: Test server started
[2026-09-17 09:28:14] SERVER: Test client connected
[2026-09-17 09:51:39] SERVER: Logger initialized successfully
[2026-09-17 09:51:39] SERVER: Server started successfully
[2026-09-17 09:51:39] SERVER: Worker thread 1 started
[2026-09-17 09:51:39] SERVER: Worker thread 2 started
[2026-09-17 09:51:39] SERVER: Worker thread 3 started
[2026-09-17 09:51:39] SERVER: Worker thread 4 started
[2026-09-17 09:54:11] SERVER: Server shutdown initiated
[2026-09-17 09:54:11] SERVER: Memory-mapped database closed
[2026-09-17 09:56:49] SERVER: Logger initialized successfully
[2026-09-17 09:56:49] SERVER: Server started successfully
[2026-09-17 09:56:49] SERVER: Worker thread 1 started
[2026-09-17 09:56:49] SERVER: Worker thread 2 started
[2026-09-17 09:56:49] SERVER: Worker thread 3 started
[2026-09-17 09:56:49] SERVER: Worker thread 4 started
[2026-09-17 09:56:57] SERVER: Client connected: 127.0.0.1
[2026-09-17 09:58:21] SERVER: Client disconnected: 127.0.0.1
[2026-09-17 09:59:41] SERVER: Client connected: 127.0.0.1
[2026-09-17 10:01:01] SERVER: Client disconnected: 127.0.0.1
[2026-09-17 10:08:00] SERVER: Client connected: 127.0.0.1
[2026-09-17 10:08:17] SERVER: Client connected: 127.0.0.1
[2026-09-17 10:11:19] SERVER: Server shutdown initiated
[2026-09-17 10:11:19] SERVER: Memory-mapped database closed
[2026-09-17 10:12:36] SERVER: Logger initialized successfully

```

```

[2026-09-23 06:20:18] SERVER: Memory-mapped database closed
[2026-09-23 15:58:52] SERVER: Logger initialized successfully
[2026-09-23 15:58:52] SERVER: Server started successfully
[2026-09-23 15:58:52] SERVER: Worker thread 1 started
[2026-09-23 15:58:52] SERVER: Worker thread 2 started
[2026-09-23 15:58:52] SERVER: Worker thread 3 started
[2026-09-23 15:58:52] SERVER: Worker thread 4 started
[2026-09-23 16:00:57] SERVER: Client connected: 127.0.0.1
[2026-09-23 16:10:03] SERVER: Logger initialized successfully
[2026-09-23 16:10:24] SERVER: Logger initialized successfully
[2026-09-23 16:11:29] SERVER: Logger initialized successfully
[2026-09-23 16:11:29] SERVER: Server started successfully
[2026-09-23 16:11:29] SERVER: Worker thread 1 started
[2026-09-23 16:11:29] SERVER: Worker thread 2 started
[2026-09-23 16:11:29] SERVER: Worker thread 4 started
[2026-09-23 16:11:29] SERVER: Worker thread 3 started
[2026-09-23 16:11:42] SERVER: Client connected: 127.0.0.1
[2026-09-23 16:13:39] SERVER: Server shutdown initiated
[2026-09-23 16:13:39] SERVER: Memory-mapped database closed
[2026-09-23 16:27:48] SERVER: Logger initialized successfully
[2026-09-23 16:27:48] SERVER: Server started successfully
[2026-09-23 16:27:48] SERVER: Worker thread 1 started
[2026-09-23 16:27:48] SERVER: Worker thread 3 started
[2026-09-23 16:27:48] SERVER: Worker thread 4 started
[2026-09-23 16:27:48] SERVER: Worker thread 2 started
[2026-09-23 16:29:13] SERVER: Client connected: 127.0.0.1
[2026-09-23 16:35:32] SERVER: Client connected: 127.0.0.1
[2026-09-23 16:36:50] SERVER: Client disconnected: 127.0.0.1
[2026-09-23 16:37:01] SERVER: Client connected: 127.0.0.1
aswitha_patel@Aswitha:~/database_client_server$ |

```

Show Configuration

```

aswitha_patel@Aswitha:~/database_client_server$ cat config.txt
PORT=8080
WORKERS=4
MAX_CLIENTS=20
LOG_LEVEL=1
aswitha_patel@Aswitha:~/database_client_server$ |

```

Demonstrate Signal Shutdown

```

aswitha_patel@Aswitha:~/database_client_server$ kill -TERM 1369
aswitha_patel@Aswitha:~/database_client_server$ |

```

In server terminal: -

```

=====
  Database Client-Server Management System
=====
Server running on port 8080
Maximum clients: 20
Worker threads: 4
Waiting for client connections...

===== STUDENT DATABASE =====
ID: 1 | Name: Aswitha | Course: CSE | Marks: 92.00 | Attendance: 95.00%
ID: 20 | Name: Alice | Course: CSE | Marks: 88.00 | Attendance: 90.00%
ID: 11 | Name: StudentB | Course: ECE | Marks: 75.00 | Attendance: 90.00%
ID: 21 | Name: Bob | Course: ECE | Marks: 92.00 | Attendance: 95.00%
ID: -1774936961 | Name: Manasvi | Course: CSE | Marks: 99.00 | Attendance: 89.00%
ID: 6 | Name: Amulya | Course: CSE | Marks: 99.00 | Attendance: 89.00%
ID: 30 | Name: Ravi | Course: CSE | Marks: 95.00 | Attendance: 95.00%
=====

===== SERVER MONITOR =====
Server PID: 1369
VmRSS:      2324 kB
Threads:      8
=====

Shutting down server...
Server stopped successfully.
aswitha_patel@Aswitha:~/database_client_server$ |

```